

Multiscale modelling of residual thermal stresses and off-axis bending in 3D braided composites

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ABSTRACT

As the failure of three-dimensional (3D) braided ceramic matrix composites (CMCs) at the structural scale are inherently influenced by their complex braid architecture at the mesoscale, a multiscale modelling method is proposed to investigate the mechanical response and damage evolution of 3D braided CMCs under combined loads induced by off-axis bending. The measured load-displacement response from off-axis three-point bend tests on 3D braided CMCs specimens for different off-axis angles agreed well with the model predictions. The study has identified the sequence and extent of various modes of damage, such as yarn breakage, matrix cracking, and yarn stripping. The multiscale model is also employed to analyze residual thermal stresses and the flexural response of 3D braided CMCs at different temperatures. The results indicate that the stiffness of the composite increases with temperature and peaks around 1000 °C. Above this temperature, the weakening of matrix-fiber interfacial bonding and the accumulation of internal damage results in a gradual decrease in the elastic modulus, with significant reductions observed above 1500 °C. These findings provide insight into the design and optimization of 3D braided CMCs destined for use in high-temperature and complex service environments.

1. Introduction

As a result of their exceptional high thermal stability, three-dimensional (3D) braided ceramic matrix composites (CMCs) have emerged as a favored material in the design of nozzle components for the aerospace industry [1–5]. In practical applications, these components are not only exposed to extreme high temperature environments but they are also subjected to complex combined mechanical loads. Under such conditions, these composites exhibit non-linear characteristics and multiple failure modes, including yarn breakage, matrix cracking and yarn stripping. The initiation and evolution of these damage modes have a direct impact on the service life and reliability of the material. The damage evolution of 3D braided CMCs structures is strongly dependent on how structural loads affect the deformation of the braided microstructure and the stresses therein.

Existing studies have mainly focused on damage development in 3D braided composites under simple loading conditions, such as tension

[6–11], compression [12–14], bending [15–17] and shear [18]. Conventional analysis methods have often employed hierarchical linear macroscopic homogenization approaches, that are unable to track nonlinear behavior. Although full-scale direct numerical simulations (DNS) can accurately capture the mechanical behavior while accounting for microstructural events, they suffer from a low computational efficiency and a prohibitive cost, particularly when simulating complex braided structures. The development of multi-scale analysis methods [19] offers a promising new approach in the analysis of braided composites. He et al. [20,21] proposed a data-driven self-consistent clustering analysis with a multiscale simulation approach, which can be used to accurately analyze the nonlinear response of braided composites under complex stress conditions. Fang et al. [22] combined the finite element method with Fast Fourier Transform (FFT) to develop a concurrent dual-scale model for a progressive damage analysis under flexural loads. Wang et al. [23] extended the FFT approach to a three-scale analysis, observing excellent damage prediction capabilities under

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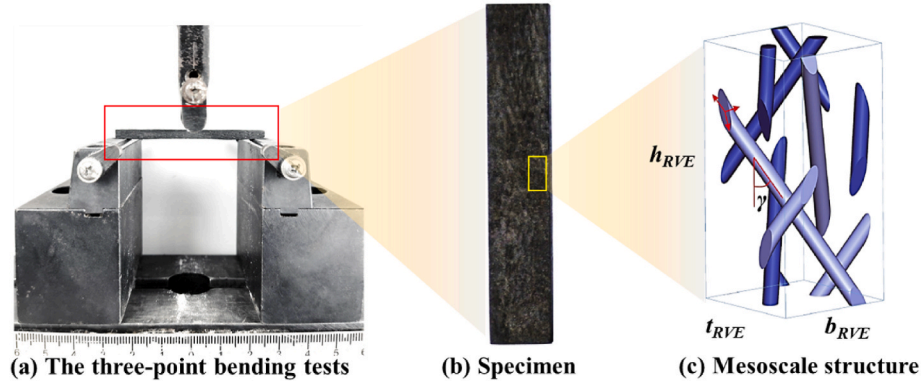


Fig. 1. Three-point bend test set-up and RVE of braided CMCs.

complex compression loads. Gao et al. [24] proposed a mechanical model for braided composites using a self-consistent clustering analysis within a multiscale framework. Although traditional concurrent multiscale methods are effective, they require an iterative interaction between the mesoscale and macroscale models, involving complex subroutine programming and *trans-scale* data exchange [25–29]. This increases the computational cost, complexity, and time, posing significant challenges for engineering applications, particularly when integrating concurrent multiscale methods within commercial finite element software.

The Direct FE² method (DFE²) is a concurrent multiscale method established to simplify the workflow and reduce computational costs associated with nested iterative schemes [30–35]. By utilizing multi-point constraints (MPCs) for the *trans-scale* kinematics, DFE² combines the computation of meso and macro models into a unified modeling framework. This enables direct and synchronized multiscale model solutions, eliminating the need for *trans-scale* data exchange.

Research into the mechanical performance and failure mechanisms of 3D braided CMCs under complex loading scenarios remains limited. An analysis of the mechanical response of 3D braided CMCs under combined loads can be used to identify the evolution of their internal damage modes and further clarify the failure mechanisms in the materials. This is important for improving the design accuracy and service reliability of such composite materials.

This study integrates a Timoshenko beam-based DFE² method with a progressive damage analysis approach to investigate the mechanical response of 3D braided CMCs. The study establishes an off-axis bending mechanical analysis model and elucidates the damage and failure mechanisms of 3D braided CMCs subjected to loading at various off-axis angles, providing clear insights into their mechanical behavior and failure processes.

2. Experiments

Three-point bending tests were performed in accordance with the ASTM C1341-06 standard [36]. Specimens with a thickness of 3 mm were machined to a final size of 60 mm in length and 9 mm in width. A span-to-thickness ratio of 16:1 was applied, as illustrated in Fig. 1. The 3D braided fiber preforms were fabricated by the China Textile Academy using T700-12K carbon fiber (Toray®, Tokyo, Japan) through a four-step braiding technique. 3D braided CMCs were obtained using the Precursor Infiltration and Pyrolysis (PIP) method [37], involving thermal oxidative crosslinking through air curing of the 3D braided preforms at 120–200 °C for 2–5 h, followed by high-temperature pyrolysis at 1000 °C to form the ceramic matrix [38].

3. Meso-structure of 3D braided composites and progressive damage model

3.1. The RVE of 3D braided CMCs

The Representative Volume Element (RVE) of the composite is shown in Fig. 1(c). The cross-section of the yarns is assumed to be elliptical, with major and minor radii a and b . The spatial geometric relationships of the RVE are given by:

$$b = \frac{\sqrt{2}}{8} b_{RVE} \quad (1)$$

$$a = \sqrt{3} b \cos \gamma \quad (2)$$

$$\tan \gamma = \sqrt{2} W_{RVE} / H_{RVE} \quad (3)$$

where b_{RVE} , t_{RVE} and h_{RVE} are the width, thickness and height of the RVE respectively. γ is the internal braiding angle.

3.2. Progressive damage model

The Chamis model is employed to estimate the stiffness and failure parameters of the yarns, as indicated in Eq. (4).

$$\begin{cases} E_{y11} = E_{f11} V_f + E_m V_m \\ E_{y22} = E_{y33} = \frac{E_m}{1 - \sqrt{V_f}(1 - E_m/E_{f22})} \\ G_{y12} = G_{y13} = \frac{G_m}{1 - \sqrt{V_f}(1 - G_m/G_{f12})} \\ G_{y23} = \frac{G_m}{1 - \sqrt{V_f}(1 - G_m/G_{f23})} \\ \nu_{y12} = \nu_{y13} = V_f \nu_{f12} + V_m \nu_m \\ \nu_{y23} = V_f \nu_{f23} + 2V_m V_m - (E_{fb2}/E_{fb1}) V_m \nu_{y12} \end{cases} \quad (4)$$

Here, E , G , ν and V represent the tension modulus, shear modulus, Poisson's ratio and volume fraction. The subscripts 1, 2, 3 denote directions normal and transverse to the yarn cross section. The subscripts y , f and m indicate yarn, carbon fiber, and matrix components, respectively.

The strength parameters of the yarns are given by Eq. (5):

Table 1
Mechanical properties of the constituent materials of the 3D braided CMCs [40].

	E_{11} (GPa)	E_{22} (GPa)	ν_{12}	ν_{23}	G_{12} (GPa)	G_{23} (GPa)	T (MPa)	C (MPa)
T700-12K	300	18.22	0.27	0.25	36.60	7.00	4900.00	1400.00
SiC	460	–	0.20	–	–	–	300.00	300.00

$$\begin{cases}
 T_{y11} = T_f (V_f + V_m E_m / E_{f11}) \\
 C_{y11} = C_f (V_f + V_m E_m / E_{f11}) \\
 T_{y22} = T_{y33} = (1 - (\sqrt{V_f} - V_f)(1 - E_m / E_{f22})) T_m \\
 C_{y22} = C_{y33} = (1 - (\sqrt{V_f} - V_f)(1 - E_m / E_{f22})) C_m \\
 S_{y12} = S_{y13} = (1 - (\sqrt{V_f} - V_f)(1 - G_m / G_{f12})) T_m \\
 S_{y23} = \sqrt{\frac{1 + T_{y22} / C_{y22}}{3 + 5 T_{y22} / C_{y22}}}
 \end{cases} \quad (5)$$

$$\mathbf{D} = \begin{bmatrix}
 d_1^2 C_{11} & d_1 d_2 C_{12} & d_1 d_3 C_{13} & 0 & 0 & 0 \\
 & d_2^2 C_{22} & d_2 d_3 C_{23} & 0 & 0 & 0 \\
 & & d_3^2 C_{33} & 0 & 0 & 0 \\
 & & & d_{12} C_{44} & 0 & 0 \\
 & & & & d_{13} C_{55} & 0 \\
 \text{sym} & & & & & d_{23} C_{66}
 \end{bmatrix} \quad (11)$$

The mechanical parameters of the constituent components of the 3D braided CMCs are summarized in Table 1.

4. Direct FE² modeling

A concurrent multiscale method for the bending performance of the 3D braided CMCs is presented. The method, Direct FE², is integrated with Timoshenko beam theory to investigate the coupling between load and deformation at the structural scale to stresses and damage evolution at the mesoscale. This approach provides computational efficiency and predictive accuracy. The *trans*-scale relationships required to establish the multiscale model are presented in brief below. Further details can be found in reference [41].

4.1. Trans-scale energy consistency

According to the Hill-Mandel condition [42], the energy density at each macroscale Gauss point must be equal to the volume average of the corresponding mesoscale RVE in a multiscale model for energy consistency. For a macroscale beam element, this can be written as:

$$F_{1t}^2 = \left(\frac{\sigma_{11}}{T_{y11}} \right)^2 + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{11} \geq 0) \quad (6)$$

The compressive tensile failure coefficient F_{1c} is:

$$F_{1c}^2 = \left(\frac{\sigma_{11}}{C_{y11}} \right)^2 + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{11} < 0) \quad (7)$$

The transverse and normal failure coefficient is:

$$\begin{aligned}
 F_{2(3)t}^2 &= \left(\frac{\sigma_{22} + \sigma_{33}}{T_{y22}} \right)^2 + \frac{\sigma_{23}^2 - \sigma_{33}\sigma_{22}}{S_{y23}^2} + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{22} + \sigma_{33} \geq 0) \\
 F_{2(3)c}^2 &= \frac{\sigma_{22} + \sigma_{33}}{C_{y22}} \left[\left(\frac{C_{y22}}{2S_{y23}} \right)^2 - 1 \right] + \left(\frac{\sigma_{22} + \sigma_{33}}{2S_{y23}} \right)^2 + \frac{\sigma_{23}^2 - \sigma_{33}\sigma_{22}}{S_{y23}^2} + \frac{\sigma_{12}^2}{S_{y12}^2} + \frac{\sigma_{13}^2}{S_{y13}^2} \geq 1, (\sigma_{22} + \sigma_{33} < 0)
 \end{aligned} \quad (8)$$

The damage criterion proposed by Christensen [40] is adopted to simulate matrix damage, with failure, F_m , defined as:

$$F_m^2 = \left(\frac{1}{T_m} - \frac{1}{C_m} \right) (\sigma_{11} + \sigma_{22} + \sigma_{33}) + \frac{1}{T_m C_m} \left\{ \frac{1}{2} [(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2] + 3(\sigma_{12}^2 + \sigma_{13}^2 + \sigma_{23}^2) \right\} \geq 1 \quad (9)$$

When the damage criteria are satisfied, the stiffness of the damaged material is reduced. The damage variables can be calculated as follows:

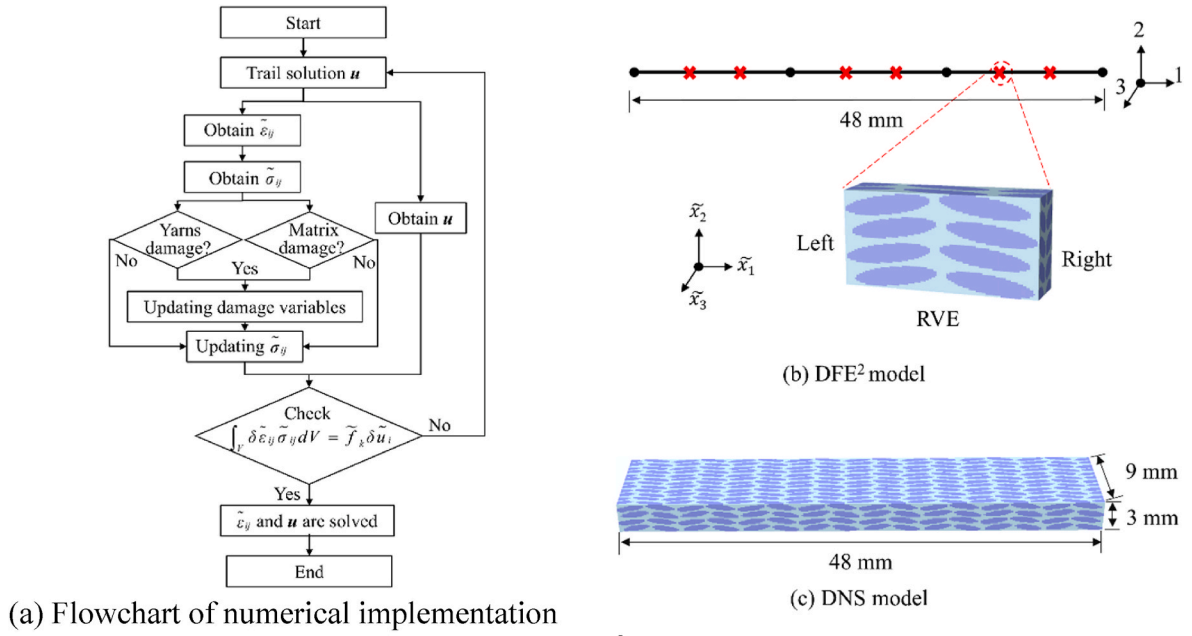
$$d_I = \min \left(1, \max \left(0, \frac{\varepsilon_I^e (\varepsilon_I - \varepsilon_I^0)}{\varepsilon_I (\varepsilon_I^e - \varepsilon_I^0)} \right) \right), I = F_{1t}, F_{1c}, F_{2t}, F_{2c}, F_{3t}, F_{3c}, F_m \quad (10)$$

where d_I , ε_I^e , ε_I , ε_I^0 are the damage variable, equivalent failure strain, equivalent strain, equivalent initial damage strain, respectively.

The stiffness matrix of the material is attenuated based on Murakami's degradation model [41].

$$\delta \tilde{W}_{int} = \sum_e \sum_\alpha \left(\int_{\tilde{V}_\alpha} \delta \tilde{\varepsilon}_{ij} \tilde{\sigma}_{ij} d\tilde{V} \right)_e = \sum_\alpha \frac{\omega_\alpha J_\alpha A_e}{\tilde{V}_\alpha} \int_{(RVE)_\alpha} \tilde{\sigma}_{ij} \delta \tilde{u}_{ij} d\tilde{V} \quad (12)$$

where e is the macroscopic element number, α a Gauss point in element e , w_α the weight of the Gauss point, J_α the Jacobian and $\tilde{V}_\alpha$ the volume of the RVE associated with the macroscale Gauss point. The superscript ' $\sim$ ' is used to indicate values calculated at the mesoscale. A_e is the cross-sectional area of the macroscale beam element. By scaling the volume of the RVE so that $V_\alpha = \omega_\alpha J_\alpha A_e$, the total strain energy of the RVEs becomes equal to the macroscale strain energy [30,43,44].

Fig. 2. Flowchart of DFE² model and DNS model.

4.2. Trans-scale kinematic constraints

To embed the RVE into the macroscale Gauss points for subsequent numerical calculations, it is necessary to relate the macroscale degrees of freedom to the degrees of freedom of the RVE by means of MPCs. From the six degrees of freedom of the macroscale beam element, the mesoscale kinematic decomposition equations with periodic boundary conditions can be written as:

$$\begin{cases} \tilde{u}_1|_R - \tilde{u}_1|_L = b_{RVE}(\nabla \tilde{u}_1^c - \tilde{x}_2 \nabla \phi_3^c + \tilde{x}_3 \nabla \phi_2^c) \\ \tilde{u}_2|_R - \tilde{u}_2|_L = b_{RVE}(\nabla \tilde{u}_2^c - \tilde{x}_3 \nabla \phi_1^c) \\ \tilde{u}_3|_R - \tilde{u}_3|_L = b_{RVE}(\nabla \tilde{u}_3^c + \tilde{x}_2 \nabla \phi_1^c) \end{cases} \quad (13)$$

where the subscripts 'R' and 'L' represent the right and left boundaries of the RVE, respectively, and b_{RVE} is the length of RVE after volume scaling.

The beam element has six degrees of freedom - displacement degrees of freedom u_1 , u_2 and u_3 along the three directions and angular degrees of freedom ϕ_1 , ϕ_2 and ϕ_3 , where the angular degrees of freedom correspond to rotations about the 1, 2 and 3 axes respectively.

4.3. Trans-scale shear angle constraints

The constraints in Eq. (13) are not sufficient to fully describe the kinematics in the mesoscale when subjected to shear deformation [45]. In order to constrain the shear angle deformation of the mesoscale, kinematic constraints are imposed by macroscale displacement moments. The average deformation of the edge nodes of the RVE is the same as that of the macroscopic beam with Gauss point rotation.

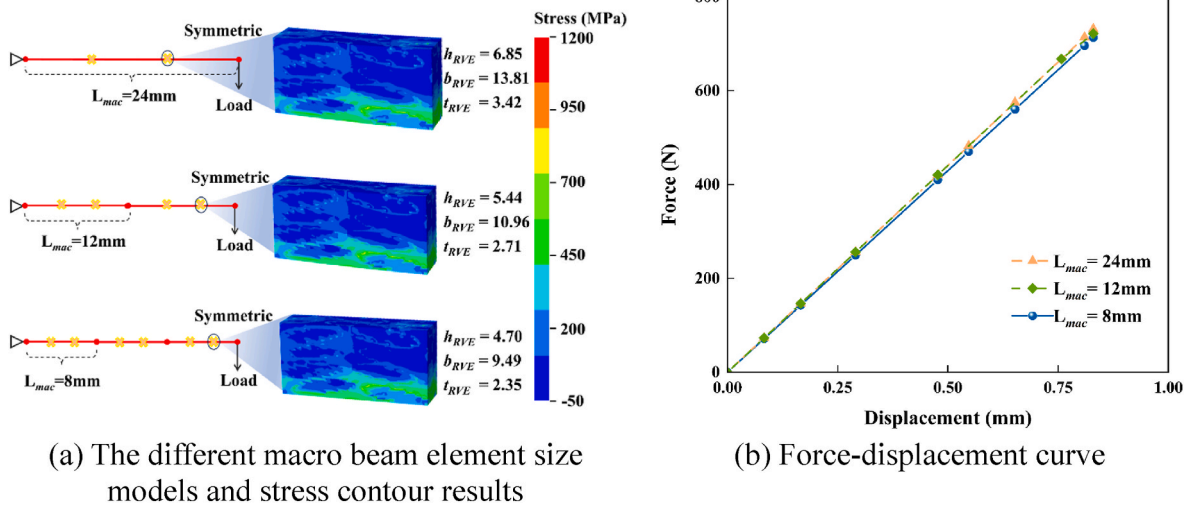
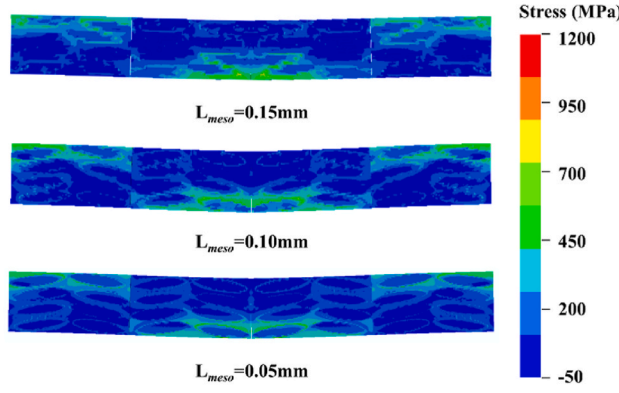
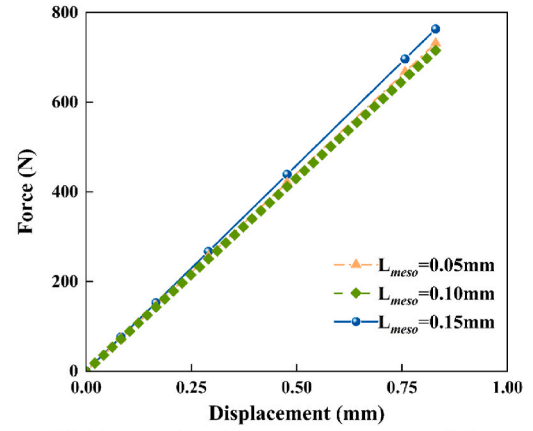


Fig. 3. Comparison of the different macro beam element size models.



(a) Mises stress contour (MPa) of the different RVE element mesh model



(b) Force-displacement curve of the different RVE element size models

Fig. 4. The different RVE element size models and stress contour results.

$$\iint \tilde{x}_3 \tilde{u}_1 d\tilde{S} = \iint \tilde{x}_3 u_1 d\tilde{S} \iint \tilde{x}_2 \tilde{u}_1 d\tilde{S} = \iint \tilde{x}_2 u_1 d\tilde{S} \quad (14)$$

The following equation can be obtained by introducing the macroscopic displacement equation into the last term of Eq. (15) and integrating.

$$\begin{aligned} \iint \tilde{x}_3 \tilde{u}_1 d\tilde{S} &= \iint \tilde{x}_3 (u_1^c - x_2 \varphi_3^c + x_3 \varphi_2^c) d\tilde{S} = \left[(\varphi_2^c) + \left(-\frac{b_{RVE}}{2} \right) \nabla \varphi_2^c \right] \left[\frac{(h_{RVE})^3 (t_{RVE})}{12} \right] \\ \iint \tilde{x}_2 \tilde{u}_1 d\tilde{S} &= \iint \tilde{x}_2 (u_2^c - x_3 \varphi_1^c) d\tilde{S} = - \left[(\varphi_3^c) + \left(-\frac{b_{RVE}}{2} \right) \nabla \varphi_3^c \right] \left[\frac{(t_{RVE})^3 (h_{RVE})}{12} \right] \end{aligned} \quad (15)$$

where b_{RVE} is the size of the RVE along the direction-1, h_{RVE} is the size of the RVE along the direction-2, t_{RVE} is the size of the RVE along the direction-3.

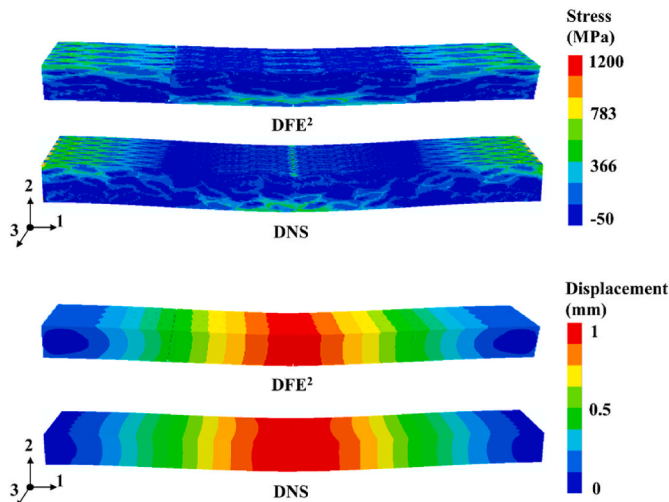


Fig. 5. Comparison of DNS and DFE² calculation results.

4.4. Implementation

Fig. 2(a) shows the flowchart of the DFE² method. With just 3 pre-processing steps, the DFE² multiscale model is implemented into the commercial finite element software ABAQUS/Standard and runs as a regular single finite element analysis (FEA) job. Details of these steps are

as follows.

- (1) Establishment of the macroscale and mesoscale model.

The geometric dimensions of the macroscale model match the bending test. The beam specimens have a span-to-thickness ratio of 16:1 and a bending span of 48 mm. The macroscale model is meshed using three-node quadratic Timoshenko beam elements (B32), with reduced integration. Macro elements are assigned near-zero material properties. The thickness of the RVEs is the same as the macrostructure. The RVEs are assigned properties described in Section 2. The mesoscale braided RVEs are periodic in the axial direction of the beam as shown in Fig. 2 (b).

- (2) Volume scaling of RVE.

To satisfy the Hill-Mandel condition, the volume of the RVE is scaled according to Eq. (12). The scaled RVEs are then fixed to the Gauss points

Table 2

Comparison of computational efficiency: CPU Time for DFE² vs. DNS models across mesh sizes.

RVE element size			
Beam mesh size	0.05 mm	0.10 mm	0.15 mm
8 mm	–	654s	–
12 mm	–	454s	–
24 mm	27s	68s	39,147s
DNS	–	1,653s	–

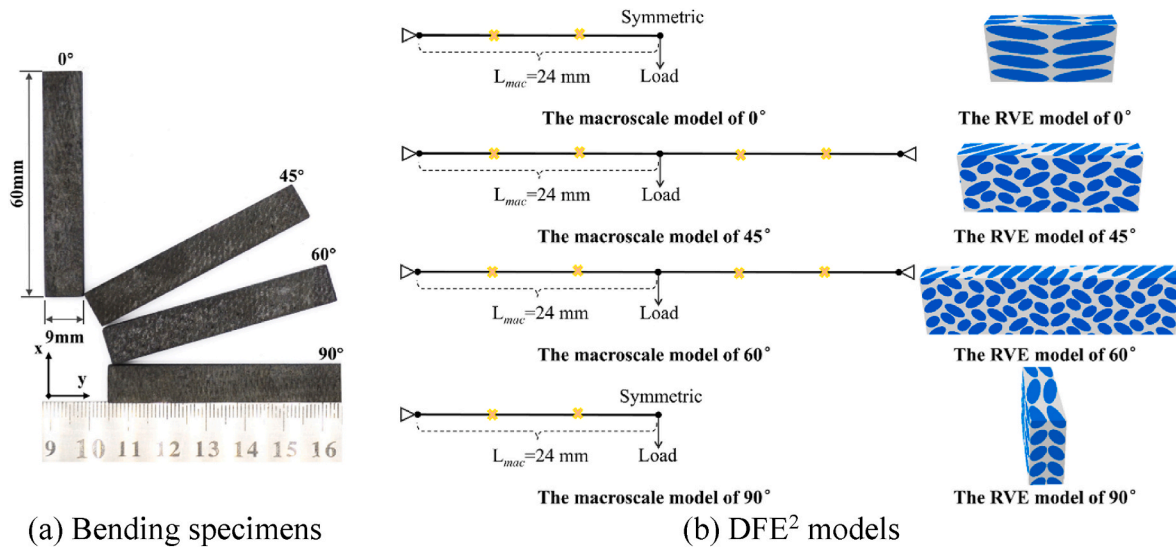


Fig. 6. Simulation model for off-axis bending of braided composites.

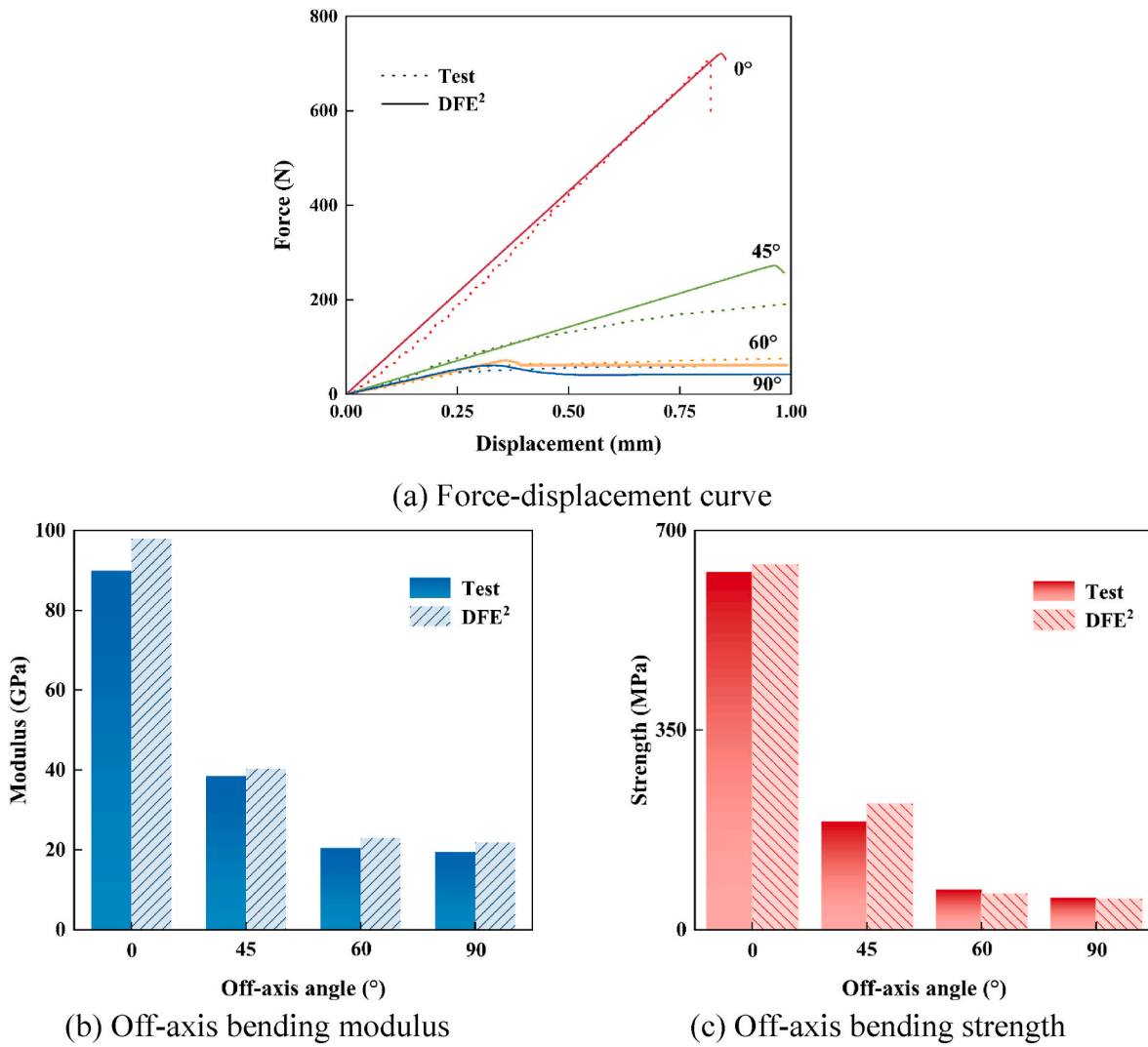


Fig. 7. Mechanical response curves under different bending loads.

Table 3
Comparison of test and DFE² results.

Off-axis angle	0°	45°	60°	90°
Modulus prediction errors (%)	8.91	4.88	12.61	12.28
Strength prediction errors (%)	2.23	14.41	9.98	3.06

of the macro beam element, aligning the centroid of each mesoscale RVE with the corresponding integration point of the macro element.

(3) Boundary and kinematic constraints.

Multi-point constraints are used to map the macroscale nodal displacements and rotations to the displacements of the RVE nodes based on Eqs. (13)–(15). These kinematic constraints are implemented via a Python script, binding the RVE end nodes of the mesoscale model to the degrees of freedom of the macroscale beam elements.

To compare the computational efficiency of the concurrent multi-scale model, a DNS model containing all braided fiber bundles is constructed as a benchmark. A displacement load is applied on the top of the beam at the mid-span to simulate bending, Fig. 2(c).

5. Mesh sensitivity analysis

A mesh sensitivity study was conducted using the model of an on-axis braided CMC beam using 3 different macroscale beam element sizes and 3 different element sizes for the RVEs. As shown in Fig. 3 (a), DFE² models were conducted using three different macro beam element sizes (24, 12, and 8 mm) to evaluate the sensitivity of the model to macroscale element size with an RVE element size of 0.1 mm. For the RVE near the center of the beam, the upper surface is subjected to a compressive stress while the lower surface is subjected to a tensile stress, consistent with the beam bending kinematics. As the macro mesh size decreases (from 24 to 8 mm), the stress contour of the RVE is similar. Fig. 3 (b) illustrates the force-displacement curves, showing their insensitivity to macro-element size for pure bending problems.

Different RVE mesh sizes (0.15, 0.10, and 0.05 mm) were then employed while maintaining a macro-element size of 24 mm. Stress contour plots were compared, as shown in Fig. 4(a). At a mesh size of 0.15 mm, the stress contours appear coarse, making it difficult to capture local variations in the mesoscale stress. As the mesh size decreases to 0.10 mm, the contour results gradually converge, and the accuracy of

the stress distribution improves significantly. When the mesh size is further reduced to 0.05 mm, the mesoscale stress is captured, exhibiting high-resolution stress distribution characteristics. Fig. 4(b) presents simulated force-displacement curves with different RVE mesh sizes. When the mesh size is 0.15 mm, the simulation results deviate increasingly from the mesh size of 0.10 and 0.05 mm. When the mesh size is smaller than 0.10 mm, the force-displacement curve does not provide much improvement in prediction accuracy.

The results of the displacement and stress predictions of DFE² (The macro beam element size = 24 mm, the RVE element size = 0.1 mm) and DNS under bending load are compared in Fig. 5. The DNS model is meshed with C3D8 elements, maintaining the same element size (0.10 mm) as the DFE² model. The results show that the two models are highly consistent in predicting displacements and stress distribution.

The balance between the number of elements and the computational cost is essential for optimizing numerical simulations. Table 2 compares the CPU time of different models, all run on a high-performance computing system (configuration: Intel(R) Xeon(R) Platinum 8375C @ 2.90 GHz, 24cores, 375 GB memory). With a beam element size of 8 mm,

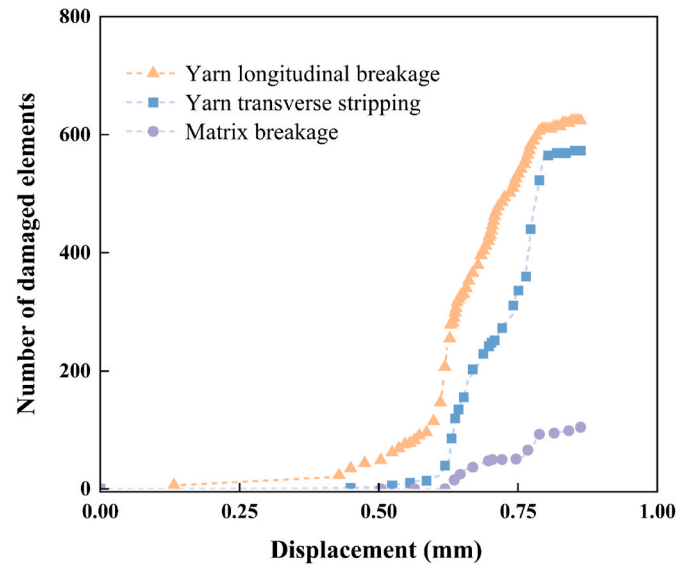


Fig. 9. Damage evolution under 0° bending load.

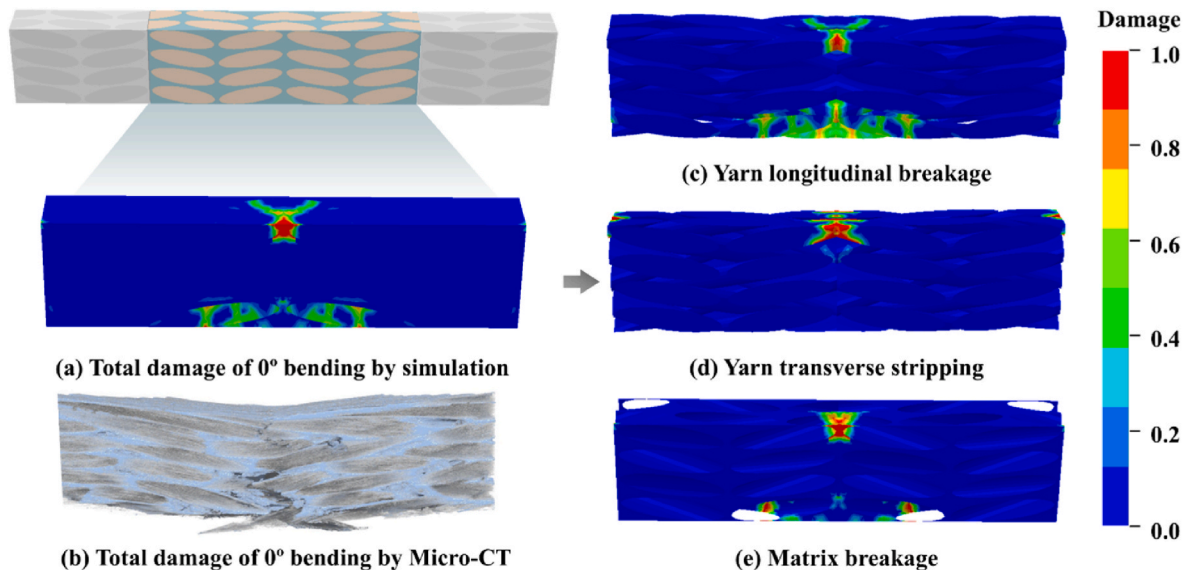


Fig. 8. Damage morphology under 0° bending.

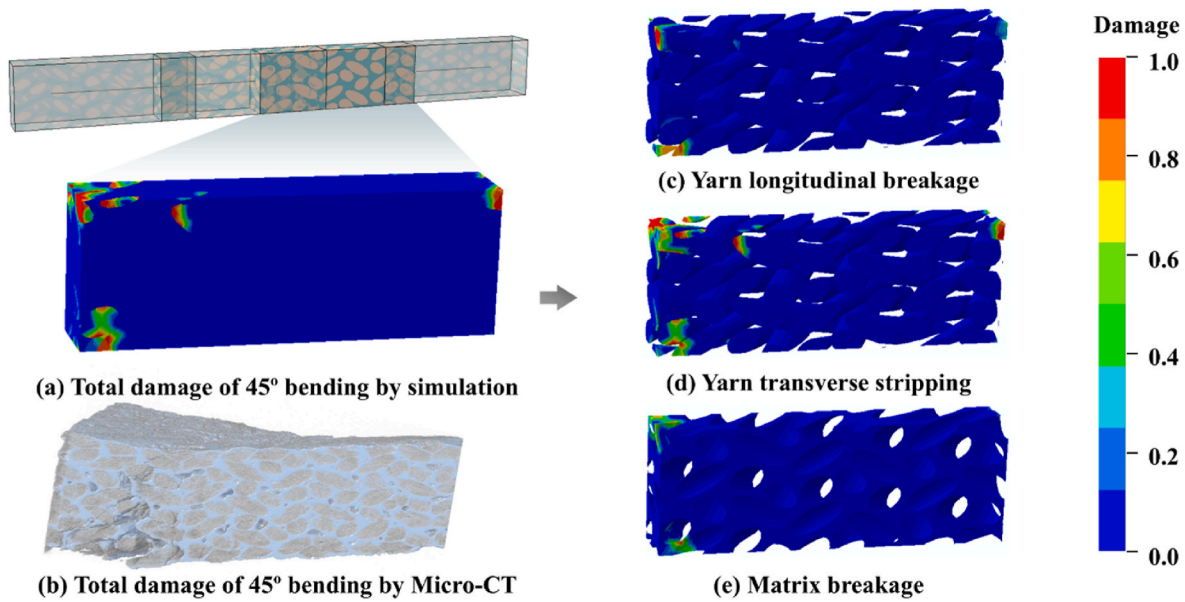


Fig. 10. Damage morphology under 45° bending.

the computation time is 654 s, while increasing the size to 24 mm reduces the computation time to 68s, an almost tenfold reduction. Based on the trade-off between accuracy and efficiency, a beam element size of 24 mm is selected for the subsequent analysis. For the different RVE element size models, a 0.15 mm mesh requires only 27s, while reducing the size to 0.10 mm increases the computation time to 68s. Further refinement of the mesh to 0.05 mm increases the computation time exponentially to 39,147s, approximately 5845 times that of the 0.10 mm mesh. A RVE mesh size of 0.10 mm is selected for further analysis. The DNS model requires 1,653s, while the DFE² model requires only 68s, reducing the computational cost by a factor of 24. This demonstrates the significant improvement in computational efficiency of the DFE² model.

6. Results and discussion

6.1. Effects of off-axis bending on 3D braided CMCs

Off-axis bending, which introduces a coupling of normal and shear stresses, effectively simulates the mixed loading conditions involving bending and torsion. Four samples with off-axis angles of 0°, 45°, 60°, and 90° were tested. Simulations of 3D braided composites with these four different off-axis angles were also performed, as shown in Fig. 6. The geometry of the macroscopic beam corresponds to the dimensions of test, with the thickness dimension aligned with the macroscopic structure. For 0° and 90° off-axis angles where the structure is symmetric, only half the beam is modeled with one end fixed and a symmetric load applied at the other end. For 45° and 60° off-axis angles, the full beam is modeled with both ends fixed and a bending load applied in the center of the beam. The RVE employs C3D4 elements with a mesh size of 0.01 mm, which was validated for data convergence in Section 4.

The calculation of bending modulus (E) and strength (S) in Eq. (16) follows ASTM C 1341-06. Fig. 7(a) compares the experimental and simulation results of the force-displacement curves for different off-axis angles. The simulation results agree well with the experiments, demonstrating the ability of the developed simulation to accurately capture the mechanical characteristics of the material. As shown in Fig. 7(b) (c), the bending modulus and strength exhibit a consistently decreasing trend with increasing off-axis angle, as observed in both the experimental and simulation results. This agreement indicates that the numerical model can reliably predict the mechanical behavior of the material under off-axis bending conditions.

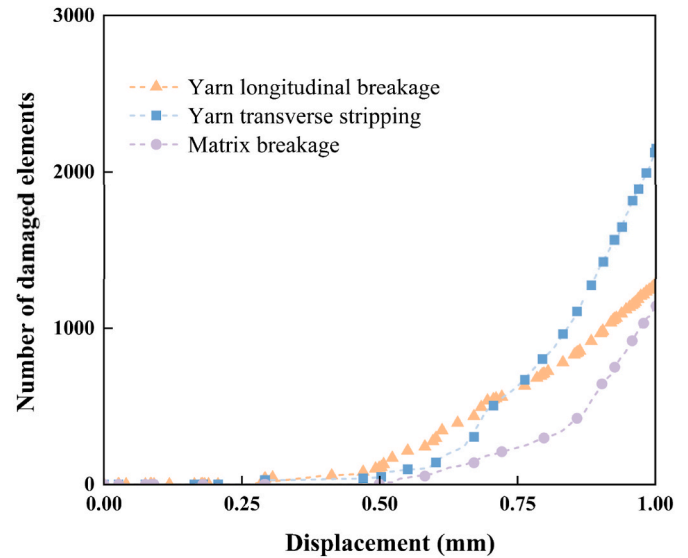


Fig. 11. Damage evolution under 45° bending load.

$$E = \frac{L^3 m}{4bd^3} S = \frac{3PL}{2bd^2} \quad (16)$$

In Eq. (16), L is the outer support span, m the slope of tangent to the initial straight-line portion of the force-displacement curve, b the width of the specimen and d the thickness of the specimen. P is the maximum force.

Table 3 shows a comparison of the experimental results with the DFE² modelling outputs. The prediction errors for both modulus and strength across all tested angles are within 15 %, confirming the accuracy and reliability of the proposed model.

Fig. 8 illustrates the damage distribution and morphology under 0° bending. The simulation results in Fig. 8(a) show the damage in the compressive zone at the top and the tensile zone at the bottom of the beam, with the damage primarily being concentrated in the tensile region of the yarn near the bottom extreme fibre. Fig. 8(b) shows the actual damage morphology reconstructed from CT scans, showing yarn fractures and crack propagation on the tensile side, indicating

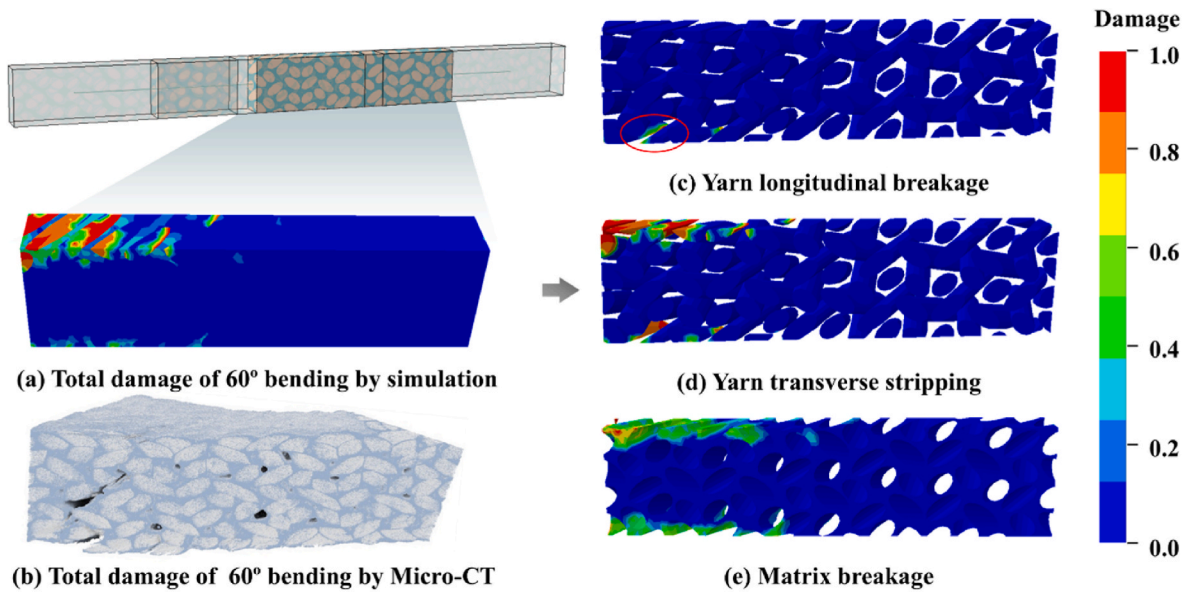


Fig. 12. Damage morphology under 60° bending.

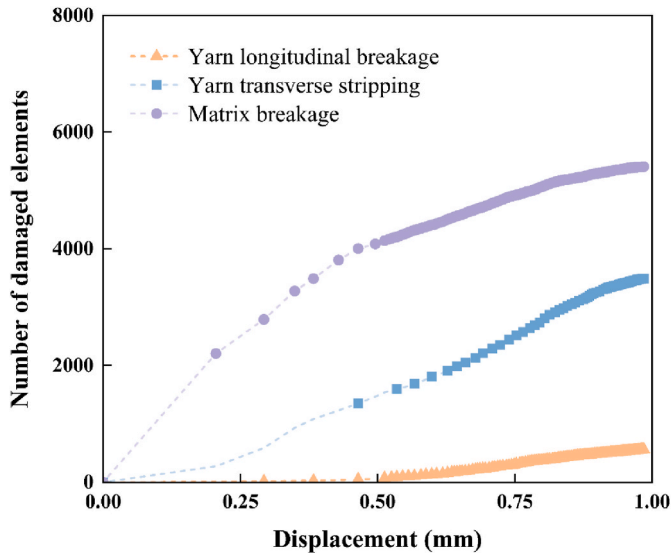


Fig. 13. Damage evolution under 60° bending load.

reasonable agreement between the experimental and simulated damage distributions. Fig. 8(c) shows longitudinal yarn fractures concentrated on the tensile side directly below the loading point. Fig. 8(d) shows transverse yarn stripping mainly along the yarn-matrix interface and Fig. 8(e) shows localized matrix fractures around the loading point and in the rear face edges of the specimen close to the yarn. The results show the typical damage mechanisms under 0° flexural loading, where longitudinal yarn fracture dominates, followed by transverse yarn stripping and minimal matrix fractures.

Fig. 9 illustrates the evolution of various damage modes in the beam subjected to 0° bending. Yarn longitudinal fracture occurs first and starts to grow at a displacement of approximately 0.25 mm and rapidly enters the accumulation phase, becoming the dominant form of damage in the initial phase. As the displacement increases to 0.5 mm, yarn transverse delamination starts to grow and develops rapidly over a short displacement range, with the cumulative damage approaching that of yarn longitudinal fracture. By contrast, matrix fracture occurs later, at a displacement of approximately 0.5 mm, and grows at a slower rate, with

a consistently lower amount of cumulative damage throughout the process. At this stage, transverse stripping of the yarn also approaches levels of saturation, while matrix fracture remains localized and contributes minimally to the overall damage. This sequence of damage evolution highlights that longitudinal fiber fracture is the primary failure mechanism in beams under 0° bending.

Fig. 10(a) shows the simulation results following 45° off-axis flexural tests, showing 45° inclined cracks, consistent with the damage distribution trends observed in the actual CT scan results shown in Fig. 10(b). Fig. 10(c) shows that longitudinal yarn fracture occurs primarily in regions where the tensile stress along the yarn direction is significant. Fig. 10(d) exhibits transverse yarn stripping concentrated at the yarn-matrix interfaces. Fig. 10(e) shows that matrix fracture is mainly located in the inter-yarn gaps within the matrix due to the release of internal stresses. Under 45° off-axis bending, yarn breakage and transverse stripping are the predominant damage modes, while matrix damage is minimal.

In the 45° off-axis bending simulation, the relative proportion of the different damage types exhibit significant variations, as shown in Fig. 11. Longitudinal yarn fracture is the first to appear, starting at a displacement of approximately 0.25 mm and gradually accumulating with loading, but its growth levelled off at displacements greater than 0.75 mm. Yarn transverse stripping grows rapidly after the displacement reaches 0.5 mm, becoming the predominant damage mode, and the number of damaged units reaches a peak when the loading approaches 1.0 mm, reflecting the significant effect of bond failure at the yarn-matrix interface. The matrix fracture damage initiated later, and only gradually appeared when the displacement was approximately 0.5 mm, with a small overall number, but showed some cumulative effects at the late stage of loading. The analysis shows that yarn transverse stripping dominates damage under 45° off-axis bending conditions.

Fig. 12(a) shows the damage evolution of the 60° bending specimen. The damage path approximately follows a 60° trajectory, aligned with the off-axis angle, and propagates along the yarn braiding direction. This is consistent with the actual damage morphology reconstructed from the CT scans as shown in Fig. 12(b). The off-axis bending load induces varying degrees of damage in the mesoscale yarns and matrix within the braided structure. In Fig. 12(c), longitudinal yarn fracture are primarily concentrated in the lower region of the RVE, with fractures propagating along the fibre direction. Fig. 12(d) shows that yarn stripping occurs predominantly in the interlaced yarn regions. Fig. 12(e) shows that brittle matrix fractures are concentrated in areas at the yarn

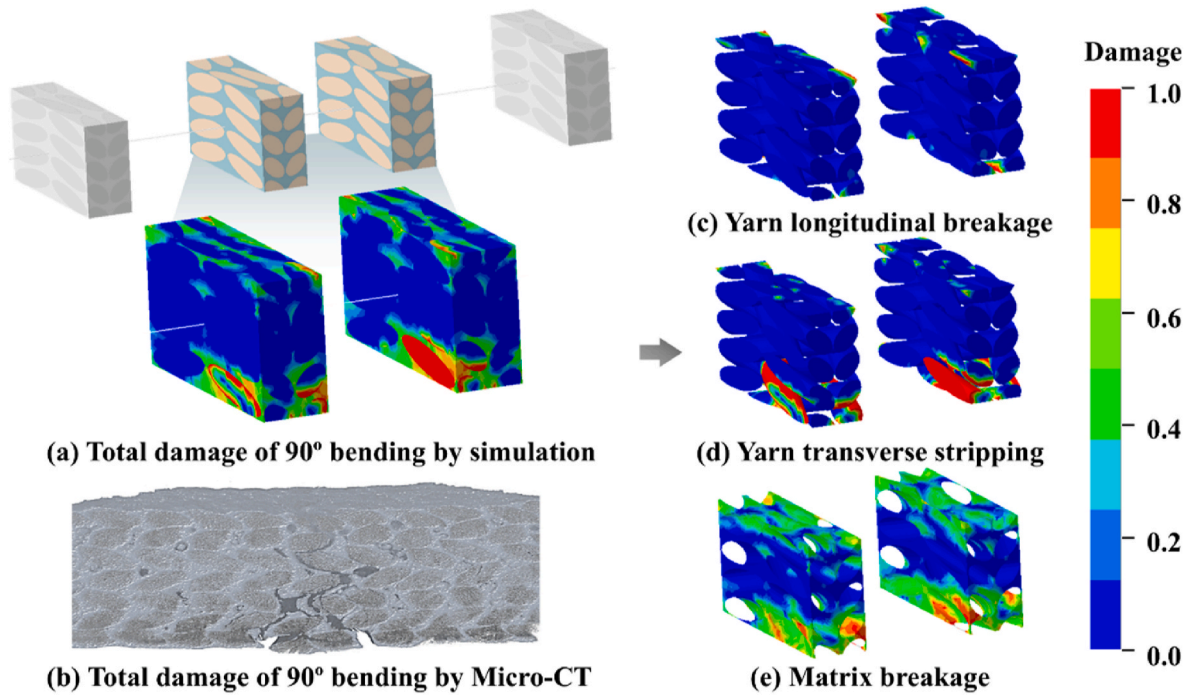


Fig. 14. Damage morphology under 90° bending.

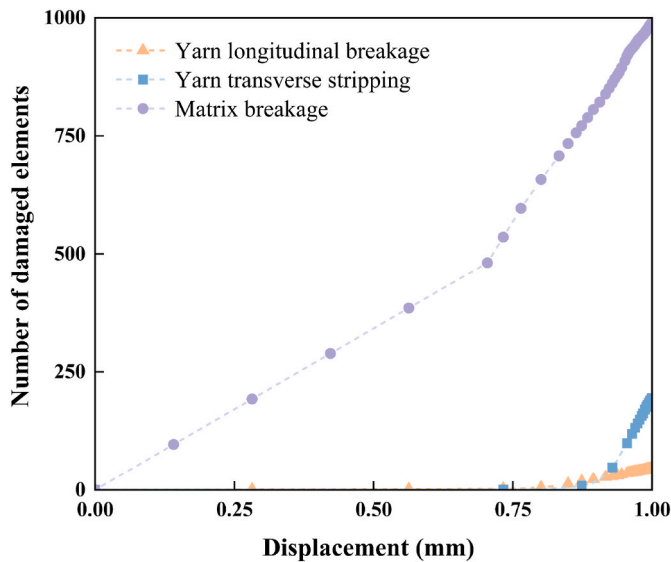


Fig. 15. Damage evolution under 90° bending load.

intersections. Under 60° off-axis bending conditions, the primary damage mechanisms are yarn stripping and matrix breakage, with the damage path closely following the off-axis angle.

Fig. 13 shows the evolution of different damage mechanisms under 60° off-axis bending. In the initial displacement range (0–0.25 mm), matrix cracks gradually appear and grow slowly, driven by localized brittle failure due to the presence of shear stresses. As the displacement increases to approximately 0.5 mm, the number of matrix cracks grows further, becoming the dominant damage mode. However, beyond 0.5 mm, the growth rate of matrix cracking begins to decrease slightly, indicating a gradual stabilization in its progression. Within the displacement range of 0.25–0.75 mm, yarn stripping begins to increase significantly, primarily due to the reduced interfacial bonding strength following matrix cracking. At this stage, the accumulation of yarn

stripping further weakens the load-bearing capacity of the yarn, accelerating the damage growth. In contrast, longitudinal yarn breakage remains at a relatively low level throughout the displacement range, with only slight increases at a displacement beyond 0.75 mm. Matrix cracking dominates the damage evolution process, followed by yarn stripping, while yarn fractures are minimal. The rapid increase in matrix cracks is the key driving factor for damage evolution, while yarn stripping and breakage represent other manifestations of failure. The brittle nature of the matrix, combined with the shear-dominated mode of off-axis bending, determines the evolution and distribution of these damage modes. The decrease in the slope of matrix cracking, reflects the interplay between the brittle matrix and the shear-dominated off-axis bending mechanism, determining the sequence and distribution of these damage modes.

Fig. 14(a) shows the simulated damage distribution under 90° bending, where damage is concentrated in the matrix near the centre of the beam and at the yarn-matrix interface. Fig. 14(b) presents the reconstructed damage morphology obtained through CT scanning, revealing interfacial delamination between the fibers and the matrix, along with localized matrix cracks. The simulated damage distribution and failure locations closely align with experimental results. Longitudinal yarn breakage is minimal under 90° bending, and is confined to localized high-stress regions, as shown in Fig. 14(c). Yarn transverse stripping is the dominant damage mode as depicted in Fig. 14(d). Matrix cracking primarily occurs near the loading points and around yarns in localized regions, indicating that the matrix undergoes significant shear and compressive deformation under 90° bending, as shown in Fig. 14(e). The damage mechanisms under 90° bending are predominantly yarn stripping and matrix cracking, with limited longitudinal yarn breakage.

The damage evolution under 90° bending, as illustrated in Fig. 15, displays a matrix-dominated characteristics. In the initial stage (displacements below 0.5 mm), the primary damage mechanism is matrix breakage, with relatively slow damage accumulation. As the displacement increases to 0.75 mm, matrix cracking enters a rapid growth phase, becoming the dominant damage mode. The cumulative damage elements of matrix cracking rise sharply, eventually reaching a maximum value of approximately 1000 elements. In contrast, yarn stripping and fracture exhibit minimal accumulation and only becomes significant at a

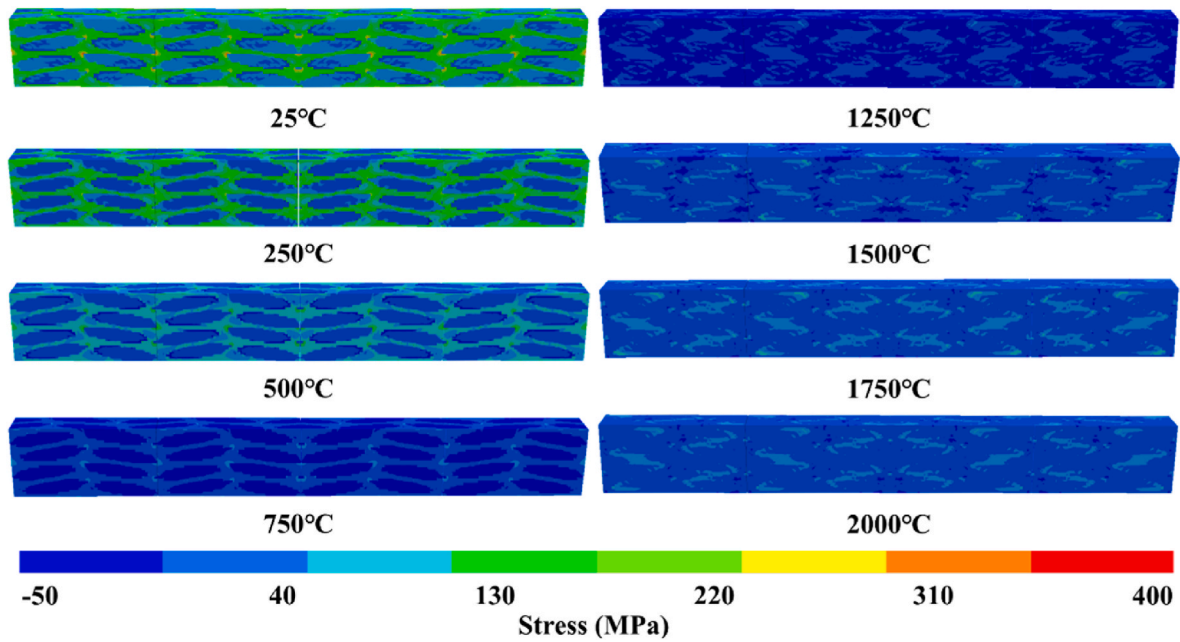


Fig. 16. Residual thermal stress (MPa) contour at different temperatures in composite beams.

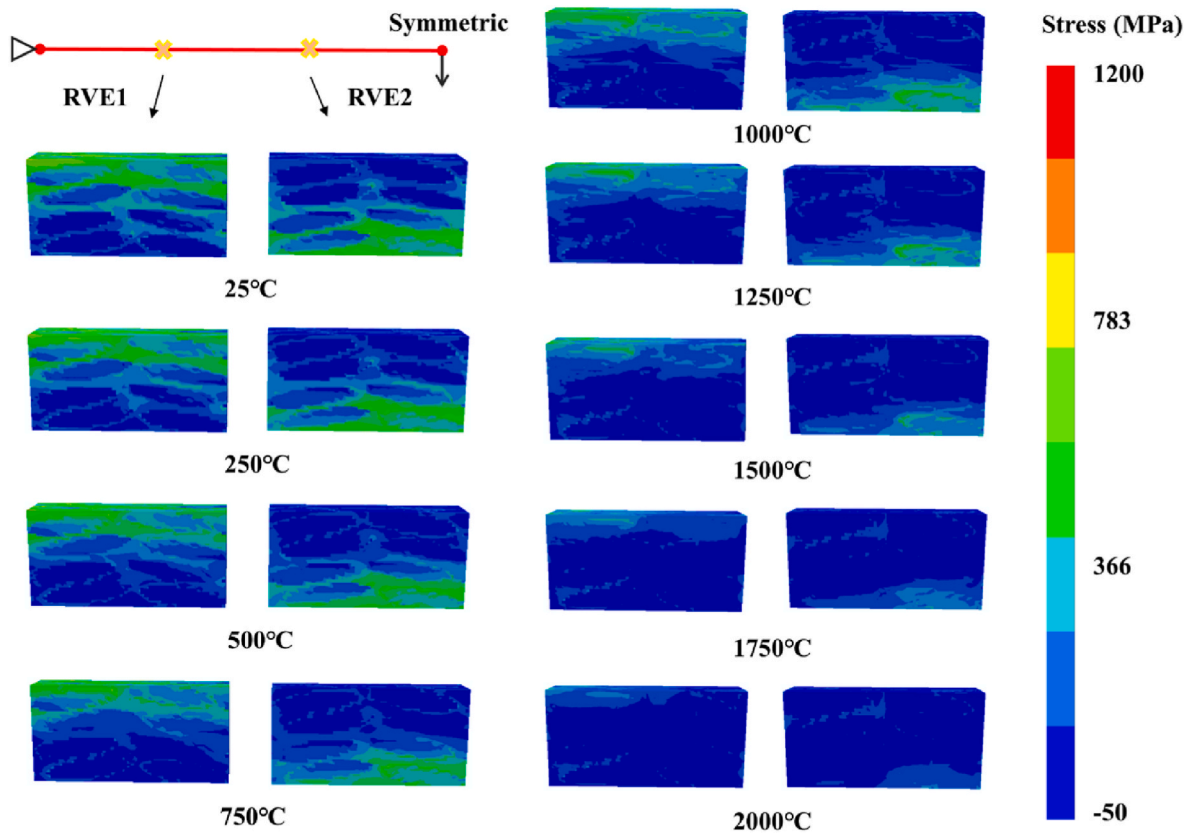


Fig. 17. Bending stress (MPa) contour plots of braided composites at different temperatures.

displacement of 1.0 mm. DFE² model effectively facilitates the quantitative analysis of damage evolution in braided composite materials under off-axis bending conditions.

6.2. Effect of thermal stress on the bending of 3D braided CMCs

3D braided CMCs are subjected to extreme thermal conditions, making the evaluation of high-temperature performance crucial. Thermal coupling analysis of 3D braided CMCs can be carried out by taking into account thermal expansion, thermal stresses and their associated

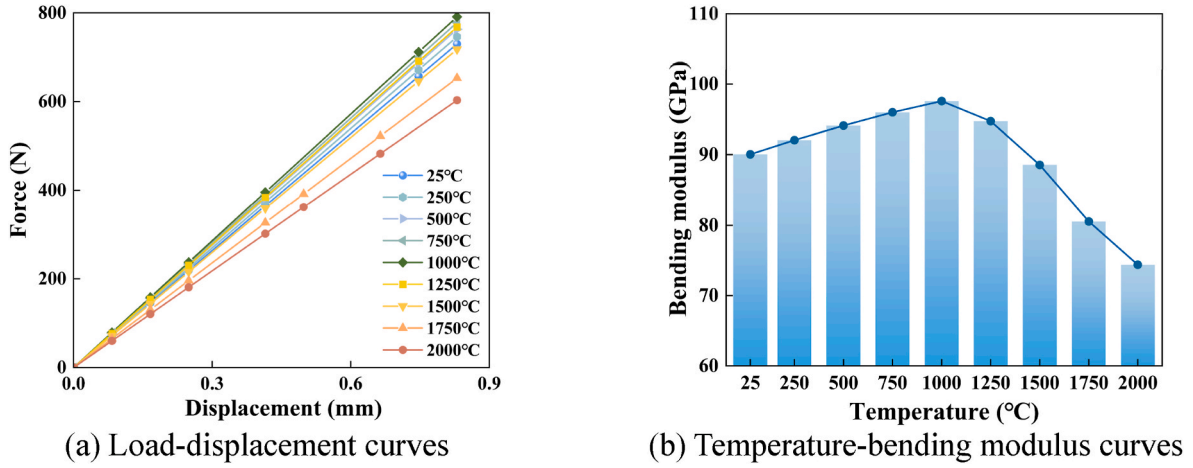


Fig. 18. Bending mechanical response of braided composites at different temperatures.

material property parameters. The thermomechanical properties of the yarn and the matrix were incorporated into the DFE2 model, operating under the assumption that the material functions are subjected to an inert atmosphere. This assumption is consistent with the experimental conditions under which the material properties are characterised. Sauder et al. [46] investigated the thermal and mechanical properties of carbon fibers at elevated temperatures. The longitudinal coefficient of thermal expansion (CTE) of carbon fibers was fitted using:

$$\alpha_f(T) = 0.200 + 8.916 \times 10^{-4}T + 8.551 \times 10^{-8}T^2, 25^\circ\text{C} \leq T < 2000^\circ\text{C} \quad (17)$$

where T is temperature ($^\circ\text{C}$).

The CTE of the SiC ceramic matrix $\alpha_m(T)$ [47] can be expressed as:

$$\alpha_m(T) = 3.244 + 1.754 \times 10^{-3}T - 5.373 \times 10^{-7}T^2, 25^\circ\text{C} \leq T < 2000^\circ\text{C} \quad (18)$$

The transverse modulus of carbon fiber is not affected by temperature but the longitudinal Young's modulus $E_f(T)$ decreases gradually with increasing temperature, which can be expressed as:

$$E_{f11}(T) = 301.9 - 3.519 \times 10^{-14}T^{4.670}, 25^\circ\text{C} \leq T < 2000^\circ\text{C} \quad (19)$$

The variation of the modulus of the ceramic matrix $E_m(T)$ with temperature is shown as follows [48]:

$$E_m(T) = \begin{cases} 459.9 - 1.632 \times 10^{-3}T^{1.387}, & 25^\circ\text{C} \leq T < 1400^\circ\text{C} \\ 422.3 - 1.224(T - 1400) + 1.144 \times 10^{-3}(T - 1400)^2, & 1400^\circ\text{C} \leq T < 2000^\circ\text{C} \end{cases} \quad (20)$$

Due to the different coefficients of thermal expansion between the carbon fiber and the ceramic matrix, thermal stresses at the interface and within the material arise during the preparation of CMC. These may lead to thermal mismatch phenomena such as cracking, warping, and peeling of the composite.

The Young's modulus of the yarn is discounted by a factor of λ from that of the fibre.

$$E_{y11}(T) = \frac{1}{2} [E_{f11}(1 - \lambda)V_f + E_m V_m(1 - K)] \quad (21)$$

A value of $\lambda = 0.3$ was adopted [49]. K is the factor to quantify the reduction in contribution of the matrix to the Young's modulus of composites. It is assumed to be proportional to the temperature

difference between the current temperature and process temperature [50]:

$$K = \frac{K_0}{|T_0 - T_p|} |T - T_p| \quad (22)$$

where T_0 ($=25^\circ\text{C}$) is room temperature, T_p ($=1000^\circ\text{C}$). For C/SiC composites, $K_0 = 0.58$.

A predefined initial temperature field (1000°C) consistent with the fabrication process was applied to the entire model. Considering the operating conditions of 3D braided ceramic matrix composites, residual thermal stresses were analyzed at temperatures of 25, 250, 500, 750, 1250, 1500, 1750, and 2000°C to evaluate the material's suitability for high-temperature applications. The internal residual stresses at different temperatures are shown in Fig. 16. In the temperature range of 25°C – 500°C , significant residual thermal stresses are observed, particularly in the fiber-matrix interface regions, where noticeable stress distributions are evident. These stresses primarily arise from the mismatch in the coefficients of thermal expansion between the yarns and the matrix, as the forming temperature decreases. At low temperatures (25°C), high residual tensile and compressive stresses develop. At 750°C , residual stresses are still present but are partially released, resulting in relatively uniform stress distribution and a slight reduction in overall stress. This indicates that, with increasing temperature, the internal residual stresses in the material undergo partial relaxation,

influenced by the initiation and closure of microcracks, which contribute to stress relief. When the temperature exceeds the manufacturing temperature, in the high-temperature range of 1250°C – 2000°C , the residual stress gradually diminishes, and stress distribution becomes more uniform. This behavior suggests that the stress state of the composite stabilizes at high temperatures. The reduction in residual stresses can be attributed to the weakened thermal mismatch effect between the matrix and yarns above the fabrication temperature. Therefore, at high temperatures close to the manufacturing temperature conditions facilitate the release of stress concentrations within the material. At this stage, internal thermal stresses are nearly eliminated, indicating that the thermal mismatch stresses are mitigated.

Fig. 17 illustrates the stress distribution in the composite samples at various temperatures ranging from 25 to 2000°C , RVE1 represents the

RVE at the beam end and RVE2 represents the RVE at the beam centre. In the low-temperature range (25–1000 °C), the mismatch in thermal expansion coefficients between the matrix and fibers contributes to thermal mismatch stress, which is particularly noticeable in regions near the supports and loading points at 25 and 250 °C. This indicates that the material retains a high level of structural integrity and strong bonding between the matrix and fibers within this temperature range. Around 1000 °C, stress becomes more uniform and decreases in intensity, as the thermal mismatch between the fibers and the matrix is reduced, and the internal residual thermal stresses are almost entirely released. In the medium-temperature range (1000–1500 °C), the stress levels slightly increase and peak in some localized areas. This is attributed to a gradual release in the residual thermal stresses within this range, where the differences in thermal expansion between the fibers and the matrix may cause localized stress concentrations without precipitating material failure. Under flexural loads, the stress distribution becomes less uniform, indicating a weakening of the bond strength between the fibers and the matrix. As the temperature rises further to the high range (1500–2000 °C), the overall stress levels decrease. Although stress concentration regions persist, the overall stress intensity diminishes compared to the medium-temperature range, suggesting that the material approaches its thermal tolerance limit.

Fig. 18(a) illustrates the load-displacement curves under three-point bending at various temperatures. The slope of the curves indicates the corresponding variation in the modulus of the composite material. Below 1000 °C, the stiffness of the material increases with increasing temperature, reaching its maximum at 1000 °C. This peak stiffness can be attributed to the release of thermal residual stresses during the fabrication process of the ceramic matrix composites. However, at temperatures above 1000 °C, the stiffness begins to decline gradually, and beyond 1500 °C, the bending modulus decreases significantly due to the internal material damage. Fig. 18(b) presents the trend in the bending stiffness of the composites at different temperatures. Below 1000 °C, the bending modulus progressively increases, which is associated with the release of thermal residual stresses and the synergistic interaction between yarns and the ceramic matrix. The closure of microcracks contributes to an enhancement in material rigidity. However, beyond 1000 °C, the degree of bonding between the matrix and fibers weakens as temperature increases, leading to a gradual reduction in bending modulus. At elevated temperatures, yarns are subjected to tensile stresses, while the brittleness of the ceramic matrix and the propagation of microcracks further diminish the modulus. Above 1500 °C, the composites exhibit a marked decrease in modulus, highlighting the adverse effects of extreme high temperatures on their mechanical performance. These results reveal the critical role of thermal expansion and thermal mismatch effects on the mechanical behavior of the composites, providing valuable insights for future material design and optimization.

Although the proposed DFE² multiscale model provides efficient and accurate predictions of the residual thermal stress behaviour of 3D braided CMCs, there are certain simplifications and assumptions. The analysis assumes an inert atmosphere and does not consider the effects of oxidative degradation at high temperatures. Furthermore, the yarn and matrix are assumed to have perfect bonding, i.e. without modelling interfacial debonding. Future work will aim to incorporate oxidation-induced damage and interfacial degradation mechanisms to enhance predictive fidelity under realistic service conditions.

7. Conclusions

Off-axis bending tests and multiscale computational modelling have been employed to study the mechanical performance and damage evolution of braided ceramic-matrix composites. The level of agreement with the experimental force-displacement plots and images from CT scan validate the simulation model. The results show that the bending modulus and strength decrease continually as the off-axis angle

increases from 0° to 90°. For 0° bending, damage is dominated by longitudinal yarn fracture, followed by transverse yarn stripping and minimal matrix fractures. Yarn transverse stripping becomes the dominant failure mode due to significant bond failure at the yarn-matrix interface in 45° bending specimens. With 60° bending specimens, matrix cracking dominates, followed by yarn stripping as damage propagated along the yarn braiding direction. As for 90° bending, matrix cracking and yarn transverse stripping are the primary failure mechanisms, with minimal longitudinal yarn breakage. The transition from yarn-dominated to matrix-dominated failure modes with increasing off-axis angle highlights the shift in stress distribution. Shear stresses play a significant role in damage evolution, particularly at high off-axis angles. The effects of residual thermal stresses were also investigated. Simulations show that below 1000 °C, the stiffness of the composites gradually increases with increasing temperature. Above 1000 °C, the bonding strength between the matrix and yarns weakens and internal damage accumulates. The elastic modulus drops significantly above 1500 °C. These findings provide valuable insights into the structural integrity of braided composites under various bending conditions, assisting their optimized design for practical applications.

CRediT authorship contribution statement

Xinyi Song: Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jin Zhou:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Kirk Ming Yeoh:** Writing – original draft, Methodology, Investigation. **Di Zhang:** Methodology. **Shenghao Zhang:** Investigation. **Karthikayen Raju:** Validation. **Xuefeng Chen:** Writing – review & editing. **Zhongwei Guan:** Writing – review & editing. **Wesley J. Cantwell:** Writing – review & editing. **Vincent Beng Chye Tan:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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